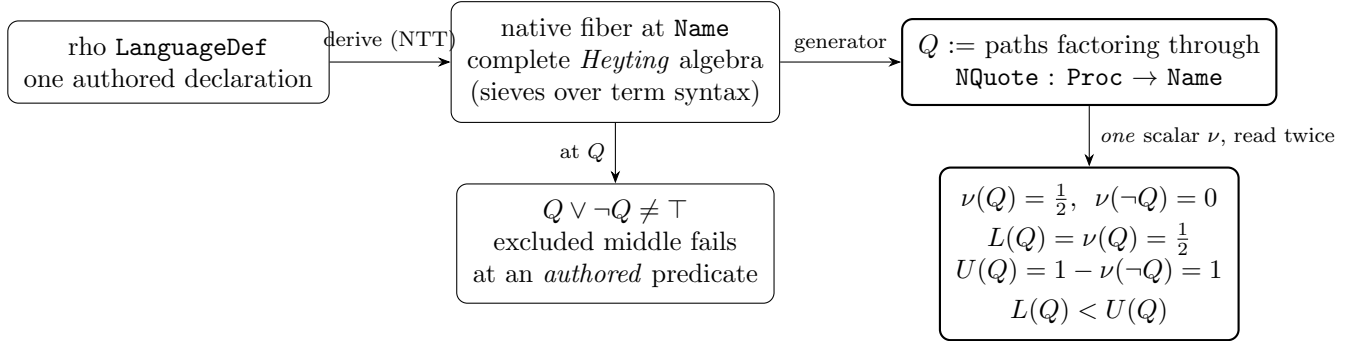


Rho's native observation logic forces two bounds

kernel-checked in Lean 4; axioms: `propext`, `Classical.choice`, `Quot.sound`



One ruler, two readings. The valuation ν is a single scalar map on the fiber. Read directly it reports *support*: $L(a) = \nu(a)$. Read through the lattice's own negation it reports *non-refutation*: $U(a) = 1 - \nu(\neg a)$. Boolean excluded middle would force $L = U$. Rho's quote-generated predicate keeps them apart — so no single point value can play both semantic roles, and a complete plausibility report about Q is the pair $(L, U) = (\frac{1}{2}, 1)$.

Why the gap appears. `NQuote` lies in Q ; the identity on `Name` does not; naturality forbids the complement from containing the identity while excluding its restriction along `NQuote`. The failure of excluded middle is the constructor geometry of quoting, nothing imported.

Lean declaration	establishes
<code>rhoQuotePredicate.excludedMiddle.ne_top</code>	$Q \vee \neg Q \neq \top$
<code>rhoNameNativeFiber.not_boolean</code>	the fiber is non-Boolean
<code>rhoCalc.nativeType.requires_distinct_probability_bounds</code>	$L(Q) = \frac{1}{2} < 1 = U(Q)$

Scope, stated exactly. The witness is structural — constructor/presheaf geometry, not COMM non-determinism; the operational $\diamond/\square$ connection is future work. The theorem concerns two independently meaningful readouts, not encoding cardinality. Knuth–Skilling fidelity is used in its published one-way form, and their quantum pair postulate is not invoked. *Which* two-component algebra to adopt (evidence pairs, amplitude–phase, ...) remains a genuine choice with genuinely different capabilities; whether a *point* value suffices for rho's native logic does not.